Project Design Phase

Solution Architecture

Date	15 February 2025
Team ID	SWTID-2026-8933
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems (e.g., granular data security and record-level entitlement in ServiceNow).
- Describe the structure, characteristics, behavior, and security evaluation workflow of the software to project stakeholders.
- Define features, development phases, and solution requirements (e.g., scripted Access Control Lists, context object checks, and role validation).
- Provide specifications according to which the solution is defined, managed, and delivered in ServiceNow Administration.

Example - Solution Architecture Diagram:

Figure 1: Architecture and data flow of Script-Controlled ACL Record Access Control in ServiceNow

Solution Architecture & Execution Flow:

• **User Request / Record Query:** End User attempts to view/read a record (e.g., Incident, HR Case, Custom Table) via ServiceNow UI or API.

• **ServiceNow Context Evaluation:** ServiceNow Access Control Engine intercepts the request and verifies table-level and field-level permissions.

• **Role & Condition Check:** ACL checks assigned User Roles (e.g., itil, snc_internal) and declarative conditions.

• **Scripted ACL Evaluation (Script Execution):** Executes server-side JavaScript utilizing the 'current' object (e.g., current.state, current.u_confidential) and 'gs.getUserID()'.

• **Access Decision (Granted / Denied):** If script sets 'answer = true', access is granted; if 'answer = false', record access is restricted/hidden.

Reference: https://docs.servicenow.com/bundle/tokyo-platform-administration/page/administer/contextual-security/concept/c_AccessControlLists.html